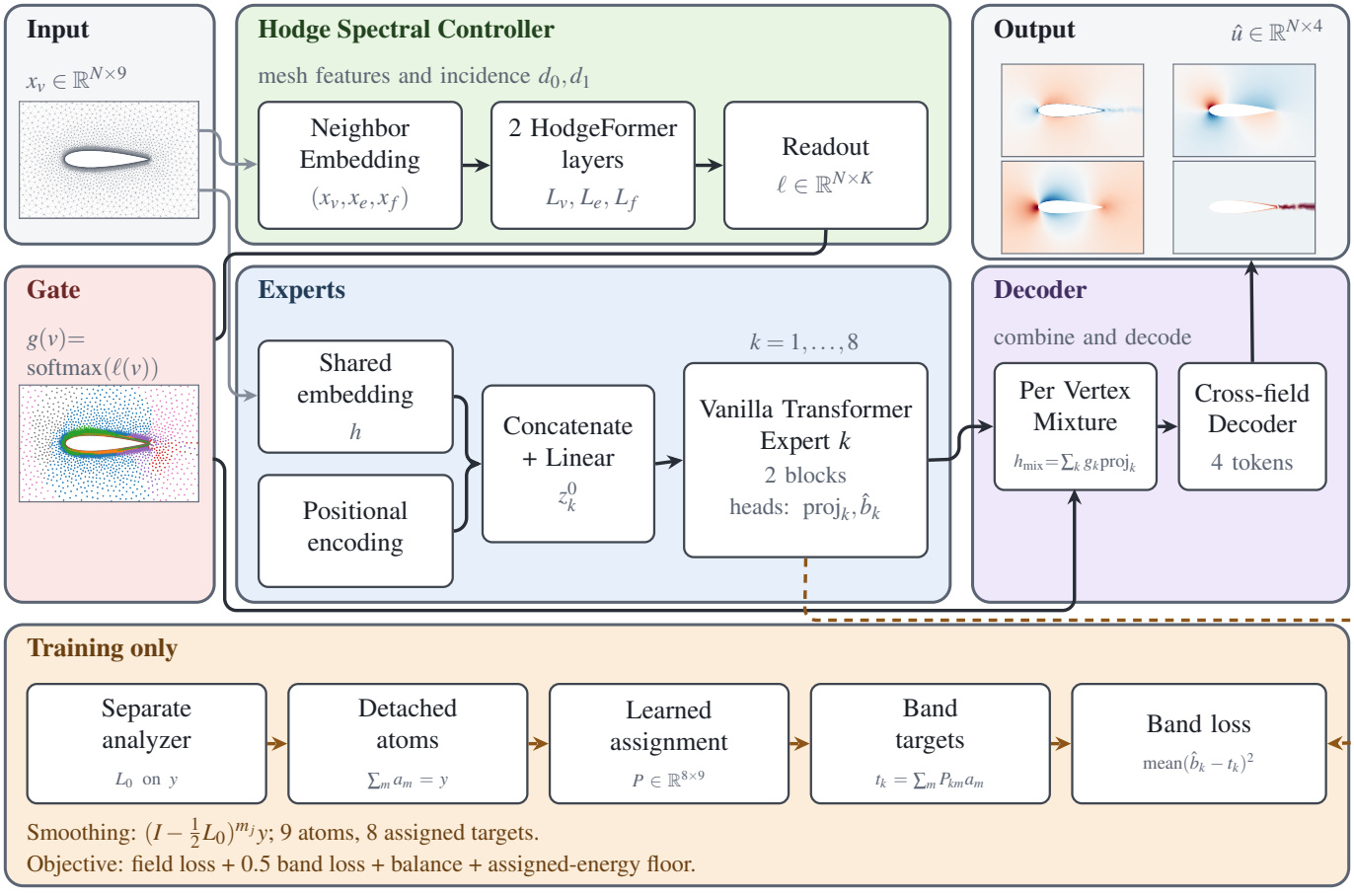

Input
$x_v \in \mathbb{R}^{N\times 9}$
Hodge Spectral Controller
mesh features and incidence d_0, d_1
Neighbor Embedding
(x_v, x_e, x_f)
2 HodgeFormer layers
L_v, L_e, L_f
Readout
$\ell \in \mathbb{R}^{N\times K}$
Output
$\hat{u} \in \mathbb{R}^{N\times 4}$
Gate
$g(v) =$ softmax$(\ell(v))$
Experts
Shared embedding
h
Positional encoding
Concatenate + Linear
z_k^0
$k = 1, \ldots, 8$
Vanilla Transformer Expert k
2 blocks
heads: $\text{proj}_k, \hat{b}_k$
Decoder
combine and decode
Per Vertex Mixture
$h_{\text{mix}} = \sum_k g_k \text{proj}_k$
Cross-field Decoder
4 tokens
Training only
Separate analyzer
L_0 on y
Detached atoms
$\sum_m a_m = y$
Learned assignment
$P \in \mathbb{R}^{8\times 9}$
Band targets
$t_k = \sum_m P_{km} a_m$
Band loss
$\text{mean}(\hat{b}_k - t_k)^2$
Smoothing: $(I - \frac{1}{2}L_0)^{m_j} y$; 9 atoms, 8 assigned targets.
Objective: field loss + 0.5 band loss + balance + assigned-energy floor.